# Ziyi Guan

ziyiguan@mit.edu • ziyiguan.github.io

## Academic appointment

**Postdoctoral Researcher**, MIT
*Hosts:* Yael Kalai, Vinod Vaikuntanathan 2026.7 – present

**Postdoctoral Researcher**, NYU
*Hosts:* Nir Bitansky, Benedikt Bünz 2026.7 – present

## Education

**Ph.D. in Computer Science**, EPFL 2021.9 – 2026.6
*Advisors:* Alessandro Chiesa, Mika Göös
*Thesis:* Kilian's Protocol and Beyond: On Constructions and Limitations of Succinct Arguments

**B.S. in Computer Science**, Carnegie Mellon University 2017.9 – 2020.12
*University Honors*

**B.S. in Mathematical Sciences**, Carnegie Mellon University 2017.9 – 2020.12
*University Honors*

## Fellowships

**SNSF Postdoc.Mobility**, Swiss National Science Foundation

## Publications

*Authors are listed alphabetically (theoretical CS convention). My name is in **bold**.*

*Conference Publications*

[BGK26] Jonathan Bootle, **Ziyi Guan**, and Christian Knabenhans. "Succinct Arguments from Lattice-Based Non-Interactive Linear Commitments". In: *Proceedings of the 24th Theory of Cryptography Conference*. TCC '26. 2026.

[BCGM26] Sarah Bordage, Alessandro Chiesa, **Ziyi Guan**, and Ignacio Manzur. "All Polynomial Generators Preserve Distance with Mutual Correlated Agreement". In: *Proceedings of the 41st Annual IEEE Conference on Computational Complexity*. CCC '26. 2026.

[CGKY26a] Alessandro Chiesa, **Ziyi Guan**, Christian Knabenhans, and Zihan Yu. "On the Fiat–Shamir Security of Succinct Arguments from Functional Commitments". In: *Proceedings of the 46th Annual International Cryptology Conference*. CRYPTO '26. 2026.

[AGGMRYS25] Yaroslav Alekseev, Mika Göös, **Ziyi Guan**, Gilbert Maystre, Artur Riazanov, Weiqiang Yuan, and Dmitry Sokolov. "Generalised Linial-Nisan Conjecture is False for DNFs". In: *Proceedings of the 40th Annual IEEE Conference on Computational Complexity*. CCC '25. 2025.

[BCG25] Annalisa Barbara, Alessandro Chiesa, and **Ziyi Guan**. "Relativized Succinct Arguments in the ROM Do Not Exist". In: *Proceedings of the 23rd Theory of Cryptography Conference*. TCC '25. 2025.

[CDDGS25] Alessandro Chiesa, Marcel Dall'Agnol, Zijing Di, **Ziyi Guan**, and Nicholas Spooner. "Quantum Rewinding for IOP-Based Succinct Arguments". In: *Proceedings of the 23rd Theory of Cryptography Conference*. TCC '25. 2025.

[GRY25] **Ziyi Guan**, Artur Riazanov, and Weiqiang Yuan. "Breaking Verifiable Delay Functions in the Random Oracle Model". In: *Proceedings of the 45th Annual International Cryptology Conference*. CRYPTO '25. 2025.

[CDGSY24] Alessandro Chiesa, Marcel Dall'Agnol, **Ziyi Guan**, Nicholas Spooner, and Eylon Yogev. "On the Security of Succinct Interactive Arguments from Vector Commitments". In: *Proceedings of the 22nd Theory of Cryptography Conference*. TCC '24. 2024.

[CGSY24] Alessandro Chiesa, **Ziyi Guan**, Shahar Samocha, and Eylon Yogev. "Security Bounds for Proof-Carrying Data from Straightline Extractors". In: *Proceedings of the 22nd Theory of Cryptography Conference*. TCC '24. 2024.

[CGY24] Alessandro Chiesa, **Ziyi Guan**, and Burcu Yildiz. "On Parallel Repetition of PCPs". In: *Proceedings of the 15th Innovations in Theoretical Computer Science Conference*. ITCS '24. 2024, 34:1–34:14.

[GHYY24] **Ziyi Guan**, Yunqi Huang, Penghui Yao, and Zekun Ye. "Quantum and Classical Communication Complexity of Permutation-Invariant Functions". In: *Proceedings of the 41st Symposium on Theoretical Aspects of Computer Science*. STACS '24. 2024, 39:1–39:19.

[GGM23] Mika Göös, **Ziyi Guan**, and Tiberiu Mosnoi. "Depth-3 Circuits for Inner Product". In: *Proceedings of the 48th International Symposium on Mathematical Foundations of Computer Science*. MFCS '23. 2023.

*Journal Publications*

[CGKY26b] Alessandro Chiesa, **Ziyi Guan**, Christian Knabenhans, and Eylon Yogev. "How to Build Succinct Arguments From Succinct Commitments". In: *Proceedings of the IEEE* (2026), pp. 1–30. DOI: 10.1109/JPROC.2026.3716313. URL: https://doi.org/10.1109/JPROC.2026.3716313.

*Manuscripts & Preprints*

[ABGH26] Hamza Abusalah, Nicholas Brandt, **Ziyi Guan**, and Kristina Hostáková. *Separating One-Way Functions from Verifiable Delay Functions*. Cryptology ePrint Archive, Paper 2026/2408. 2026. URL: https://eprint.iacr.org/2026/2408.

[BCG26] Benedikt Bünz, Jessica Chen, and **Ziyi Guan**. *It Takes Two: Proofs of Work for Fiat–Shamir*. Cryptology ePrint Archive, Paper 2026/2326. 2026. URL: https://eprint.iacr.org/2026/2326.

[CGMV26] Alessandro Chiesa, **Ziyi Guan**, Ignacio Manzur, and Thomas Vidick. *Succinctness Requires Probabilistic Checking in the Quantum World*. Manuscript. 2026.

[CGPS26] Alessandro Chiesa, **Ziyi Guan**, Omer Paneth, and Nicholas Spooner. *Time-Space Tradeoffs for Probabilistic Proofs*. Cryptology ePrint Archive, Paper 2026/2406. 2026. URL: https://eprint.iacr.org/2026/2406.

[CGWW26] Alessandro Chiesa, **Ziyi Guan**, William Wang, and Yuetian Wu. *The Cost of Rewinding for Succinct Arguments: Strict-Time Barriers and Expected-Time Security*. Cryptology ePrint Archive, Paper 2026/2209. 2026. URL: https://eprint.iacr.org/2026/2209.

[CGY26] Alessandro Chiesa, **Ziyi Guan**, and Burcu Yıldız. *Can AI Oversight Be Zero Knowledge?* arXiv:2610.01995. 2026. URL: https://arxiv.org/abs/2610.01995.

[Woo+26] David P. Woodruff et al. *Accelerating Scientific Research with Gemini: Case Studies and Common Techniques*. 2026. arXiv: 2602.03837 [cs.CL]. URL: https://arxiv.org/abs/2602.03837.

[GY25] **Ziyi Guan** and Eylon Yogev. *Micali's SNARG, Function Vector Commitments, and Fiat–Shamir*. Cryptology ePrint Archive, Paper 2025/2328. 2025. URL: https://eprint.iacr.org/2025/2328.

[CDGS23] Alessandro Chiesa, Marcel Dall'Agnol, **Ziyi Guan**, and Nicholas Spooner. *On the Security of Succinct Interactive Arguments from Vector Commitments*. Cryptology ePrint Archive, Paper 2023/1737. 2023. URL: https://eprint.iacr.org/2023/1737.

[BCGL22] Jonathan Bootle, Alessandro Chiesa, **Ziyi Guan**, and Siqi Liu. *Linear-Time Probabilistic Proofs with Sublinear Verification for Algebraic Automata Over Every Field*. Cryptology ePrint Archive, Paper 2022/1056. 2022. URL: https://eprint.iacr.org/2022/1056.

## Invited Talks

*Cryptographic Proof Systems for Scalable AI Oversight*
- Crypto & Security Seminar, New York University — Oct 2026

*The Cost of Rewinding for Succinct Arguments: Strict-Time Barriers and Expected-Time Security*
- Northeastern University — Oct 2026

*Time-Space Tradeoffs for Probabilistic Proofs*
- Seminar, Columbia University — September 2026
- ZKProof 8, Rome — May 2026

*Succinctness Requires Probabilistic Checking in the Quantum World*
- Seminar, Stanford University — August 2026
- Information-Theoretic Cryptography, UCSB — August 2026
- Seminar, Sorbonne Université — June 2026

*On the Security of Succinct Arguments from Probabilistic Proofs*
- Workshop on Zero-Knowledge, Succinct Proofs and Symmetric Cryptography, TU Vienna — Feb 2026
- Security Seminar, Boston University — Sep 2025
- CyLab Crypto Seminar, Carnegie Mellon University — Aug 2025
- Aarhus University — Jun 2025
- ETH Zurich — May 2025

*Relativized Succinct Arguments in the ROM Do Not Exist*
- TCC 2025, Aarhus — Dec 2025
- Swiss Crypto Day, EPFL — Oct 2025
- ZKProof 7, Sofia — Mar 2025

*Breaking Verifiable Delay Functions in the Random Oracle Model*
- Crypto & Security Seminar, New York University — Sep 2025
- Cryptography and Information Security Seminar, MIT — Sep 2025
- Theory Seminar, Cornell University — Aug 2025
- Crypto 2025, UCSB — Aug 2025
- NordiCrypt Summer 2025, Aarhus University — Jun 2025

*Quantum Rewinding for IOP-Based Succinct Arguments*
- Paris ZK Day, ENS — Jun 2025
- Workshop on the Mathematics of Post-Quantum Cryptography, University of Zurich — Jun 2025
- Bocconi University — May 2025

*Untangling the Security of Kilian's Protocol: Upper and Lower Bounds*
- TCC 2024, Bocconi University — Dec 2024

*Security Bounds for Proof-Carrying Data from Straightline Extractors*

- ZKProof 7, Sofia — Mar 2025
- TCC 2024, Bocconi University — Dec 2024
- Swiss Crypto Day 2024, University of St. Gallen — Sep 2024
- CrossFyre 2024 (affiliated event of Eurocrypt 2024), ETH Zurich — May 2024
- Crypto Seminar, Carnegie Mellon University — Mar 2024

*On the Security of Succinct Interactive Arguments from Vector Commitments*

- Crypto Reading Group, New York University — Feb 2024
- CYS Research Seminar, King's College London — Feb 2024
- Crypto Seminar, Bar-Ilan University — Jan 2024

*On Parallel Repetition of PCPs*

- Algorithms and Complexity Seminar, University of Cambridge — Feb 2024
- Bocconi Seminar, Bocconi University — Jan 2024
- Theory Seminar, Nanjing University — Nov 2023

*Depth-3 Circuits for Inner Product*

- MFCS 2023, Bordeaux INP — Aug 2023

## Teaching

*EPFL*

Co-instructor • Theory of Computation — Spring 2026
Head Teaching Assistant • Theory of Computation — Spring 2025
Head Teaching Assistant • Theory of Computation — Spring 2024
Teaching Assistant • Theory of Computation — Spring 2023
Teaching Assistant • Foundations of Probabilistic Proofs — Fall 2023
Teaching Assistant • Foundations of Probabilistic Proofs — Fall 2022
Teaching Assistant • Foundations of Probabilistic Proofs — Spring 2022

*SLMath (formerly MSRI)*

Teaching Assistant • Foundations and Frontiers of Probabilistic Proofs — Summer 2023

*Carnegie Mellon University*

Teaching Assistant • Great Ideas in Theoretical Computer Science — Fall 2020
Tutor • Great Ideas in Theoretical Computer Science — Spring 2020
Teaching Assistant • Continuous Time Finance — Spring 2020
Teaching Assistant • Discrete Time Finance — Fall 2019
Teaching Assistant • Integration and Approximation — Spring 2019

## Professional Service

*Program Committee*

Asiacrypt 2026

*External Reviewer*

SODA 2027, Crypto 2026, Eurocrypt 2026, STOC 2026, LATINCRYPT 2025, TCC 2025, Crypto 2025, ICALP 2025, Eurocrypt 2025, QIP 2025, Eurocrypt 2024, TCC 2023, CCC 2023